

**MEDIATED COMPLIANCE:
ARCHITECTURAL CONSTRAINTS ON THE OPERABILITY
OF INTERNATIONAL HUMANITARIAN LAW OBLIGATIONS
IN AI-MEDIATED OPERATIONAL ENVIRONMENTS**

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ABSTRACT

The relationship between artificial intelligence systems and International Humanitarian Law has been extensively analysed in terms of whether AI systems may themselves comply with humanitarian obligations, whether autonomous weapons systems may lawfully be deployed, and whether meaningful human control may be preserved across the operational chain. This paper proposes a distinct analytical category that may complement existing scholarship.

The category proposed is mediated compliance: the condition under which the practical operability of an International Humanitarian Law obligation depends on technical infrastructure designed and procured by actors distinct from the human decision-maker who bears the formal obligation. Where sensor fusion, automated classification, and algorithmic prioritisation mediate the information that reaches the human operator, architectural choices made upstream of the engagement decision may determine which compliance-relevant signal classes the operator is in a position to perceive.

This paper develops the concept in five dimensions. First, it situates mediated compliance within existing humanitarian law obligations and demonstrates that the analytical separation between architecture and obligation may be increasingly difficult to sustain. Second, it proposes a taxonomy of architectural mediation forms—sensor filtering, taxonomic exclusion, threshold calibration, and priority weighting—and illustrates their compounding through a concrete operational scenario. Third, it conducts a comparative analysis with analogous doctrines in other legal orders, including deliberate indifference in United States constitutional and tort law, *Organisationsverschulden* in German civil and criminal law, *responsabilità amministrativa degli enti* under Italian Legislative Decree 231/2001, and corporate criminal liability under Article 102 of the Swiss Penal Code. Fourth, it examines the implications of mediated compliance for state attribution under the Articles on Responsibility of States for Internationally Wrongful Acts, particularly where operational architecture is supplied by private contractors and where the procuring or hosting state holds a special position under International Humanitarian Law. Fifth, it considers the operational and procurement implications, including the institutional position of Switzerland as depositary of the Geneva Conventions and host state of the International Committee of the Red Cross.

The paper does not argue that current international law explicitly recognises mediated compliance as a binding doctrine. It advances the narrower proposition that the existing framework of humanitarian obligations, command responsibility doctrine, and state responsibility law may already contain the analytical resources necessary to address architectural mediation, and that recognising mediated compliance as a distinct analytical category may clarify ongoing debates in autonomous systems governance, defence procurement, and international legal accountability.

I. INTRODUCTION

The legal analysis of artificial intelligence systems in operational and military-adjacent environments has, to date, proceeded primarily along three tracks. The first concerns whether autonomous weapons systems may comply with International Humanitarian Law as a general matter. The second examines whether meaningful human control may serve as an operational compliance criterion. The third investigates whether specific safeguard categories may perform compliance-relevant functions within operational AI architectures.

Each of these tracks has produced substantial scholarship. Each also operates within a shared assumption: that the locus of compliance analysis is the human decision-maker, supported or constrained by technical systems whose architecture remains analytically external to the legal obligation itself. Under this assumption, the obligation rests on the human; the architecture is the environment in which the obligation is discharged.

This paper proposes that the assumption may require qualification. As operational environments become increasingly mediated by algorithmic classification, automated sensor fusion, probabilistic targeting, and machine-assisted prioritisation, the practical operability of a humanitarian obligation may depend, in significant part, on the architectural choices that determine what information reaches the human decision-maker, in what form, and with what salience. The architecture does not replace the human obligation. It conditions the informational substrate on which the obligation must be discharged.

The condition under which the practical operability of a legal obligation depends on technical infrastructure designed and procured by actors distinct from the human decision-maker who bears the formal obligation may be described as mediated compliance. Under conditions of mediated compliance, the analytical separation between architecture and obligation may become difficult to sustain. The architecture functions as a constitutive condition of the compliance environment, not merely as a contextual one.

This paper develops the concept of mediated compliance in five steps. Section II situates the concept within existing humanitarian law obligations and develops its analytical structure. Section III proposes a taxonomy of architectural mediation forms relevant to humanitarian compliance, illustrated through a concrete operational scenario and presented in tabular form. Section IV conducts a comparative analysis with analogous doctrines in domestic legal orders, examining whether existing legal frameworks contain analytical resources that may inform the development of mediated compliance as an international legal category, with particular attention to the Swiss corporate criminal liability regime under Article 102 of the Penal Code. Section V examines the implications for state responsibility, including the special institutional position of Switzerland as depositary of the Geneva Conventions and host state of the International Committee of the Red Cross. Section VI considers the operational and procurement implications. Section VII identifies the limitations of the argument.

The paper advances no claim that current international law explicitly mandates recognition of mediated compliance as a binding doctrine. The argument is descriptive-analytical: it proposes that the existing framework of humanitarian and state responsibility law may already contain the analytical resources necessary to address architectural mediation, and that articulating mediated compliance as a distinct category may clarify ongoing scholarly and policy debates.

II. THE ANALYTICAL STRUCTURE OF MEDIATED COMPLIANCE

A. The Objection from Formal Allocation

A potential objection to the framework proposed in this paper warrants direct engagement at the outset. It may be argued that humanitarian obligations attach to the human combatant who must recognise *hors de combat* status, distinguish between combatants and civilians, and apply feasible precautions in attack. Under this view, system design choices remain analytically separate from the legal obligation, which continues to rest exclusively on the human decision-maker. The architecture provides inputs; the human makes decisions; the obligation governs the decision.

This objection, while structurally coherent, may underestimate the operational reality of contemporary AI-mediated decision environments. In systems where sensor fusion, automated classification, and algorithmic prioritisation mediate the information that reaches the human operator, the architectural filtering of input data occurs prior to the human recognition act. The combatant does not perceive the operational environment directly. The combatant perceives a representation of that environment constructed by the system architecture.

Where such an architecture is designed in a manner that systematically omits compliance-relevant signal classes—whether through training data exclusion, classification taxonomy design, threshold calibration, or priority weighting—the practical capacity of the human operator to perform the recognition act on which the legal obligation depends may be foreclosed before the obligation is engaged. The legal obligation remains, formally, on the human decision-maker. The operational substrate on which that obligation depends has, however, been altered by architectural choices made upstream of the engagement decision.

B. The Constitutive Function of Architecture

The principle of feasible precaution under Article 57 of Additional Protocol I is particularly relevant in this context. Feasible precaution is not a static standard tied exclusively to the moment of engagement decision. It encompasses the full sequence of choices that determine what is operationally feasible at that moment, including procurement and design choices that determine which information classes the system is capable of preserving and presenting.

Where compliance-relevant signal classes are technically achievable—assessed against verifiable criteria including Technology Readiness Levels, sensor specifications, latency thresholds, and the availability of annotated training datasets—and are nonetheless systematically excluded at the architectural level, the feasibility envelope available to the human operator is correspondingly reduced. The architectural choice does not merely constrain the operator. It defines the practical scope of what feasible precaution can mean in the operational context that the architecture has constructed.

This analysis does not assign legal obligations to AI systems as such. The architecture remains a technical artefact, not a legal subject. The analytical move is different: the obligations of human decision-makers, and of commanders under Article 28 of Additional Protocol I and Article 28 of the Rome Statute, may extend to the procurement and design decisions that shape the informational substrate on which downstream compliance depends. The architecture is not external to the compliance analysis. It is constitutive of the operational conditions under which compliance is, or is not, achievable.

C. The Distribution of the Locus of Decision

A consequence of the analytical structure proposed above is that the locus of compliance-relevant decision-making becomes distributed across the procurement and deployment chain. Traditional International Humanitarian Law analysis, developed in pre-algorithmic operational environments, generally locates the relevant decision at the moment of engagement. The combatant identifies, distinguishes, applies precaution, and engages. The decision is temporally compact and personally attributable.

In AI-mediated operational environments, this temporal and personal compactness may dissolve. The relevant decisions are made at multiple points: at the architectural design stage, where signal classes are defined and classification taxonomies are constructed; at the training

stage, where annotated datasets determine which patterns the system learns to recognise; at the calibration stage, where thresholds determine the precision-recall trade-offs that shape system output; at the procurement stage, where acquiring authorities select systems with specific architectural properties; at the deployment stage, where commanders authorise operational use; and at the engagement stage, where the human operator acts on the system output.

Each of these decision points may contribute to the operational conditions under which the formal humanitarian obligation is, or is not, capable of being discharged. The doctrine of mediated compliance recognises this distribution and proposes that compliance analysis may need to engage with the full sequence of architectural decisions, rather than focusing exclusively on the terminal engagement moment.

D. The Customary Foundation

The proposition that architectural choices may be compliance-relevant draws on customary International Humanitarian Law obligations that are themselves well established. Rule 15 of the ICRC Customary IHL Study codifies the principle of precaution in attack as customary law binding on all parties to a conflict regardless of treaty ratification. Rule 16 establishes the customary obligation to do everything feasible to verify that targets are military objectives. Rule 47 establishes the customary prohibition on attacking persons recognised as hors de combat.

None of these rules specifies a particular architectural implementation. None mandates any specific technology. Their relevance to mediated compliance analysis arises from a different consideration: each rule presupposes that the operational actor is in a position to perform the cognitive act on which the rule depends—verification, recognition, distinction. Where architectural choices systematically remove the operational actor from that position, the customary obligation does not disappear. It becomes operationally inoperable. The doctrine of mediated compliance proposes that such operational inoperability is itself a legally relevant phenomenon, even where no single actor has explicitly violated any specific rule.

III. A TAXONOMY OF ARCHITECTURAL MEDIATION

The analytical category of mediated compliance gains operational utility when the forms of architectural mediation that may engage it are specified. This Section proposes a four-fold

taxonomy of architectural mediation forms relevant to humanitarian compliance: sensor filtering, taxonomic exclusion, threshold calibration, and priority weighting. The taxonomy is intended as an analytical reference rather than an exhaustive catalogue. Operational systems may exhibit multiple mediation forms simultaneously, and the boundaries between categories are not always sharp.

A. Sensor Filtering

Sensor filtering refers to the architectural choice of which physical signal modalities the system is designed to capture and process. An optical sensor suite optimised for thermal contrast may detect human presence reliably but may not capture gesture-level information relevant to surrender recognition. An acoustic sensor optimised for vehicle signatures may not capture verbal surrender expressions. A radar-based system may register movement and stationary objects without capturing the visual or postural information that may indicate hors de combat status.

The legal relevance of sensor filtering arises where the omitted modality carries compliance-relevant information that is technically capturable through alternative or supplementary sensor configurations. Where such alternative configurations are technically achievable and operationally feasible within the system parameters, the choice to deploy a sensor configuration that excludes the compliance-relevant modality is an architectural decision that may engage the mediated compliance framework.

Sensor filtering operates at the physical input layer of the system. It is, in a literal sense, prior to any subsequent processing. Information that is not captured cannot be classified, weighted, or escalated. The architectural choice at this layer therefore has a particularly strong constitutive effect on the downstream feasibility envelope.

B. Taxonomic Exclusion

Taxonomic exclusion refers to the architectural choice of which categories the system is designed to recognise. A classification system may receive sensor data containing surrender-relevant signal patterns and nonetheless fail to identify them as such, because the classification taxonomy does not include a category to which those patterns may be assigned. The information is captured, but it is not assigned compliance-relevant meaning within the system's representational scheme.

Taxonomic exclusion is particularly significant because it operates at the level of representational ontology. A system that categorises detected human figures as either "hostile" or "non-hostile" may not have a representational structure capable of expressing the intermediate state of "potentially hors de combat" or "surrender-ambiguous." Sensor data that would, under a richer taxonomy, be classified into a compliance-relevant intermediate state is forced into one of the available binary categories. The information is preserved in the raw data; it is lost in the representational output.

This form of mediation is closely related to the design of training datasets. Where annotated training data does not include examples of surrender-relevant configurations labelled as such, the resulting classification model cannot reliably identify those configurations regardless of how much capacity the underlying architecture possesses. The taxonomic exclusion is partly an artefact of training data composition and partly an artefact of the categorical structure that the dataset is built to support.

C. Threshold Calibration

Threshold calibration refers to the architectural choice of precision-recall trade-offs in classification systems. A surrender recognition system that requires high confidence before classifying an observation as surrender-relevant may exhibit low recall: many genuine surrender configurations may be classified as non-surrender or as uncertain. A system calibrated with lower confidence thresholds may exhibit higher recall but also higher false positive rates.

The legal relevance of threshold calibration arises because the precision-recall trade-off is not a neutral technical parameter. It encodes an implicit prioritisation of error types. A system calibrated to minimise false positives at the expense of recall may, in operational deployment, generate higher rates of failure to identify genuine hors de combat conditions. The architectural choice to calibrate in this manner may itself be compliance-relevant, particularly where the operational consequence of false negatives is engagement of persons protected under Article 41 of Additional Protocol I.

Threshold calibration interacts with the feasible precaution analysis under Article 57 of Additional Protocol I. Where calibration is biased toward operational efficiency at the cost of recall in compliance-relevant signal categories, the system architecture may reduce the operability of

feasible precaution obligations without explicitly excluding any signal class. The mediation operates not through omission but through the quantitative configuration of detection sensitivity.

D. Priority Weighting

Priority weighting refers to the architectural choice of how detected information is ordered, presented, and emphasised in the operational output delivered to the human decision-maker. A decision support system that detects surrender-relevant signals and includes them in its output, but ranks them below threat-relevant signals in the salience hierarchy, may functionally suppress compliance-relevant information even where it is technically present in the system's processing.

This form of mediation is particularly significant in high-tempo operational environments where the human operator's attention is necessarily selective. The architectural decision about what is presented first, what is highlighted, and what is relegated to secondary displays may determine which information actually informs the engagement decision. Compliance-relevant information that is technically available but operationally subordinated may be functionally absent from the decision-making process.

Priority weighting is closely related to the user interface design of operational systems. The cognitive ergonomics of how information is presented to the operator are not external to the compliance analysis. They are part of the architectural mediation that determines what feasibly informs the engagement decision.

E. The Compounding of Mediation Forms

In operational systems, the four mediation forms described above frequently compound rather than operating in isolation. A sensor configuration that does not capture gesture information (sensor filtering) feeds a classification system whose taxonomy lacks a surrender-ambiguous category (taxonomic exclusion), calibrated with high precision thresholds biased toward minimising false positives (threshold calibration), with a user interface that displays threat scores prominently and de-escalation indicators in secondary panels (priority weighting). Each mediation form independently reduces the operability of feasible precaution obligations. Together, they may constitute an architectural environment in which compliance-relevant recognition is foreclosed at multiple successive layers.

The compounding effect is analytically significant because it suggests that mediated compliance analysis cannot proceed by examining each mediation form in isolation. The operational consequence depends on the interaction of mediation forms across the architectural stack. An audit framework that examines only sensor specifications, or only classification taxonomies, or only threshold settings, may miss the compounding effect that emerges from their combination.

F. Illustrative Compounding Scenario

The following scenario is offered as an analytical construct to illustrate how the four mediation forms may compound in operational deployment. It does not describe any existing system and is not intended to evaluate any specific procurement decision.

An intelligence, surveillance, and reconnaissance platform is procured for deployment in a densely populated operational environment characterised by frequent civilian presence and ambiguous combatant status. The platform integrates a multi-sensor suite optimised primarily for thermal contrast detection at extended ranges. The thermal sensor configuration is selected because it performs reliably under low-visibility conditions and supports the operational tempo required by the mission profile. The architectural decision to optimise for thermal contrast, however, results in degraded resolution for the postural and gestural information that may signal cessation of hostilities. This is the first mediation form: sensor filtering.

The sensor data feeds a classification system trained on annotated datasets developed for force protection applications. The training data labels detected human figures as either "armed," "unarmed," or "unknown." The taxonomy does not include intermediate categories such as "surrender-ambiguous," "recently disengaged," or "posture inconsistent with active engagement." A combatant who has discarded a visible weapon and raised hands is classified as "unarmed"—correctly, in the literal sense—but the classification does not preserve information about the transition from prior armed status or the gestural indicators that may engage Article 41 protections. This is the second mediation form: taxonomic exclusion.

The classification system is calibrated with high confidence thresholds. The calibration rationale, documented in procurement specifications, prioritises minimisation of false positive classifications of civilians as combatants—itself a compliance-relevant design choice consistent

with the principle of distinction. The same threshold calibration, however, applied symmetrically to the detection of surrender-relevant gestural patterns, results in low recall for ambiguous surrender signals. Gestural configurations that do not reach the confidence threshold are not flagged as compliance-relevant; they are classified within the existing taxonomy and presented without surrender-specific salience. This is the third mediation form: threshold calibration.

The operator interface displays the classification outputs alongside threat assessment scores derived from the system's threat prioritisation logic. The threat assessment scores are presented prominently in the primary display field. Secondary classification attributes, including any ambiguity indicators that may have been generated by the system, are accessible through a subordinate display layer that the operator can query but that is not surfaced by default. Under operational tempo, the operator's attention is drawn to the primary display. This is the fourth mediation form: priority weighting.

In aggregate, the operator perceives an operational environment in which threats are highlighted and ambiguity is suppressed. The architectural design does not violate any explicit rule. Each design choice can be defended on operational grounds. The compounded effect, however, is an informational environment in which the obligations under Articles 41 and 57 of Additional Protocol I may be inoperable as a practical matter, not because the operator has failed to apply them but because the architectural mediation has constructed an operational reality in which the recognition act on which those obligations depend is foreclosed. The mediated compliance framework proposes that this aggregate consequence is itself a legally relevant phenomenon, distinct from the assessment of any individual design choice in isolation.

G. Summary of the Taxonomy

The four mediation forms described in this Section may be summarised as a structural taxonomy. Table 1 presents the analytical features of each form, the architectural layer at which it operates, and the principal compliance-relevant consequence.

Table 1: Forms of Architectural Mediation in AI-Mediated Operational Systems

Mediation Form	Architectural Layer	Mechanism	Compliance-Relevant Consequence
Sensor filtering	Physical input layer	Selection of signal modalities; omission of modalities carrying compliance-relevant information	Information not captured cannot be classified, weighted, or escalated downstream
Taxonomic exclusion	Representational ontology	Absence of compliance-relevant categories in classification taxonomy and training data	Information captured but not assigned compliance-relevant meaning in system output
Threshold calibration	Quantitative configuration	Precision-recall trade-off biased against recall in compliance-relevant signal classes	Genuine compliance-relevant signals fail to reach classification confidence threshold
Priority weighting	Operator interface and salience design	Ordering, emphasis, and presentation hierarchy of detected information	Compliance-relevant information technically present but functionally suppressed under operational tempo

The taxonomy supports the analytical move proposed in Section II: that the locus of compliance-relevant decision is distributed across the procurement and deployment chain. Each row of Table 1 identifies a distinct decision point at which architectural choices may engage the mediated compliance framework. The taxonomy also supports the comparative doctrinal analysis developed in Section IV by providing a structured object for legal characterisation.

IV. COMPARATIVE DOCTRINAL FOUNDATIONS

The analytical category of mediated compliance proposed in this paper is novel in its application to International Humanitarian Law and AI-mediated operational systems. The underlying analytical move, however—recognising that organisational and architectural conditions may engage legal accountability beyond the formal locus of decision—is not without precedent in other legal orders. This Section examines three doctrinal analogues from domestic legal systems that may inform the development of mediated compliance as an international legal category: the doctrine of deliberate indifference in United States constitutional and tort law; the

doctrine of Organisationsverschulden in German civil and criminal law; and the regime of responsabilità amministrativa degli enti under Italian Legislative Decree 231/2001. None of these doctrines is directly transposable to International Humanitarian Law. Each, however, illustrates how domestic legal systems have engaged with the analytical problem of distributed responsibility in conditions where formal allocation of obligation does not capture the operational reality of harm causation.

A. Deliberate Indifference in United States Law

The doctrine of deliberate indifference has developed across several domains of United States law, including Eighth Amendment jurisprudence on prison conditions, Fourteenth Amendment substantive due process analysis, and municipal liability under Section 1983 of Title 42 of the United States Code. In each domain, the doctrine addresses circumstances in which legal responsibility may attach to actors who did not themselves directly cause harm but whose conduct or organisational decisions created conditions under which harm became foreseeable and substantially likely.

In the municipal liability context, the Supreme Court has held that a local government may incur liability under Section 1983 where its policies or training programmes exhibit deliberate indifference to constitutional rights. The doctrinal structure articulated in *Canton v. Harris* and refined in subsequent decisions requires identification of a pattern of inadequacy that creates a foreseeable risk of constitutional violation, and recognition by the relevant policymaking authority that the inadequacy is likely to result in such violation. The legal accountability attaches not to the individual actor who directly causes the violation but to the organisational structure that conditioned the violation.

The analogy to mediated compliance is partial but instructive. The deliberate indifference doctrine recognises that organisational architecture—training programmes, supervisory structures, resource allocation—may create conditions of foreseeable rights violation even where each individual actor performs within the discretion afforded by the organisational design. The legal response is to attach accountability to the organisational level at which the architectural conditions were established.

Mediated compliance shares the analytical move of recognising that conditions established upstream of the harm-causing decision may engage legal accountability. It differs in two principal respects. First, mediated compliance does not require a showing of subjective awareness on the part of the organisational decision-maker; the analysis is closer to an objective feasibility test. Second, mediated compliance operates in an international rather than a domestic legal framework, with consequent differences in the available accountability mechanisms.

B. Organisationsverschulden in German Law

The German doctrine of Organisationsverschulden—organisational fault—has developed primarily in the context of civil liability for organisational entities, particularly under sections 31 and 831 of the Bürgerliches Gesetzbuch. The doctrine recognises that an organisational entity may be liable for harm caused by its operations where the organisational structure itself was inadequately designed to prevent the foreseeable harm. The fault lies not in the conduct of any specific individual within the organisation but in the design of the organisation itself.

The doctrine has been applied in cases involving medical institutions, industrial enterprises, and public authorities. In each context, the analytical question is whether the organisational design exhibited a structural inadequacy that made the type of harm in question reasonably foreseeable, and whether reasonable organisational measures could have addressed the inadequacy. Where both conditions are met, Organisationsverschulden may be established without identification of individual misconduct.

The analogy to mediated compliance is closer here than in the United States doctrine, for two reasons. First, Organisationsverschulden focuses on the design of the organisational system rather than on subjective states of awareness. The inquiry is structural rather than psychological. Second, the doctrine explicitly recognises that the inadequacy of organisational design may engage legal accountability even where each individual actor within the organisation performed competently within the scope of their role.

Translated to the AI-mediated operational context, the Organisationsverschulden analytical structure suggests that the inadequacy of architectural design—the failure to include compliance-relevant signal classes, the calibration of thresholds in ways that systematically suppress recall in protected categories, the priority weighting that subordinates de-escalation information—may be

analytically analogous to the structural inadequacy that engages organisational fault. The doctrine does not directly apply, but its analytical architecture is relevant.

C. Responsabilità Amministrativa degli Enti under Italian Legislative Decree 231/2001

Italian Legislative Decree 231 of 8 June 2001 introduced a regime of administrative responsibility for legal entities in connection with certain offences committed in their interest or to their advantage. The regime is formally administrative rather than criminal, but it operates within the criminal procedure framework and imposes substantial sanctions including fines, prohibitions on activities, and confiscation. The conceptual structure of the regime is particularly relevant to mediated compliance analysis.

Under Article 6 of the Decree, an entity may be exonerated from liability where it can demonstrate that it had adopted and effectively implemented an organisational model adequate to prevent the type of offence in question, that supervision of the model had been entrusted to an autonomous body, and that the offence was committed through fraudulent circumvention of the model. The structure of the regime thus operationalises the concept that organisational architecture is itself the locus of accountability: the entity is liable not for the conduct of individual actors but for the inadequacy of the organisational model that conditioned the conduct.

The Italian model is particularly illuminating because it explicitly couples organisational accountability with documented architectural specifications. The organisational model must be written, must be verifiable, must include monitoring mechanisms, and must be subject to independent supervision. Liability is engaged where these architectural specifications are inadequate to the foreseeable risks generated by the entity's activities. The legal accountability is anchored in documented design choices, not in unverifiable cultural attributes of the organisation.

Translated to the mediated compliance context, the Italian regime suggests that legal accountability for architectural inadequacy may be operationalised through documentation requirements: written architectural specifications, verifiable feasibility assessments, independent monitoring of architectural choices, and explicit identification of the compliance-relevant risks that the architecture is designed to address. The documentation itself becomes the locus of accountability analysis.

D. Corporate Criminal Liability under Article 102 of the Swiss Penal Code

The Swiss regime of corporate criminal liability under Article 102 of the Penal Code, in force since 2003, presents a structurally distinctive analytical model that warrants particular attention in connection with mediated compliance. The regime distinguishes between two forms of corporate criminal liability. Under paragraph 1, an undertaking is liable for any felony or misdemeanour committed in the exercise of its commercial activities and in accordance with its objectives where the offence cannot be attributed to a specific natural person owing to inadequate organisation. Under paragraph 2, an undertaking is liable for an enumerated list of specific offences—including criminal organisation, financing of terrorism, money laundering, and corruption—where the undertaking has failed to take all reasonable organisational measures required to prevent such offences, irrespective of any natural person's criminal liability.

Three features of the Swiss regime are particularly relevant to the mediated compliance framework. First, the regime explicitly anchors liability in organisational inadequacy. Paragraph 1 conditions liability on the inability to attribute the offence to a natural person owing to inadequate organisation; paragraph 2 conditions liability on the failure to adopt reasonable organisational measures. In both cases, the locus of legal accountability is the organisational architecture itself, not the conduct of any specific individual. This structural feature aligns precisely with the mediated compliance analytical move of locating accountability in architectural conditions.

Second, the scope of paragraph 1 is broad. Unlike the closed catalogue in paragraph 2, paragraph 1 applies to any felony or misdemeanour committed in the exercise of the undertaking's commercial activities. The Swiss implementation of international humanitarian law in Articles 264 through 264j of the Penal Code, enacted in 2011 to operationalise the Geneva Conventions and the Rome Statute, defines war crimes and crimes against humanity as felonies under Swiss criminal law. Under the monist Swiss legal tradition, customary international law and ratified treaties are automatically incorporated into the domestic legal order. The result is a structural configuration in which an undertaking operating in Switzerland, or under Swiss jurisdiction, that contributes through inadequate organisational architecture to the commission of war crimes—where the natural person attribution fails for organisational reasons—falls within the literal scope of Article 102 paragraph 1.

Third, the Swiss regime applies to undertakings under both private and public law, with the exception of local public authorities. Article 102 paragraph 4 of the Penal Code defines undertakings broadly to include legal entities under private law, legal entities under public law other than territorial corporations, companies, and sole proprietorships. Defence contractors organised under Swiss commercial law, whether wholly Swiss-owned or operating Swiss subsidiaries, fall within the scope of the regime.

The analytical implications for the mediated compliance framework are substantial. A defence contractor supplying AI-mediated operational architecture to a state procuring authority, where the architecture exhibits the mediation forms described in Section III, and where deployment of the architecture contributes to operational outcomes constituting war crimes under Articles 264c through 264j of the Penal Code, may—in principle and subject to all the evidentiary and procedural conditions of Swiss criminal procedure—engage the literal scope of Article 102 paragraph 1. The analytical condition that activates the regime is the same one that the mediated compliance framework identifies as legally relevant: inadequate organisational architecture that conditions the commission of the predicate offence.

This analytical observation is not a claim that any Swiss defence contractor is presently exposed to Article 102 prosecution. The Swiss corporate criminal liability regime has been applied principally in connection with corruption, money laundering, and related economic offences, and prosecutorial practice in connection with war crimes liability of corporate actors has not yet developed. The observation is structural: the legal architecture for analysing corporate accountability for organisational contributions to humanitarian violations exists within Swiss domestic law, and the mediated compliance framework articulates the analytical structure that such accountability would engage.

The Swiss regime also illustrates a feature of the comparative analysis that warrants emphasis. Across the four doctrinal analogues examined in this Section—the United States deliberate indifference doctrine, the German Organisationsverschulden, the Italian responsabilità amministrativa degli enti, and the Swiss Article 102 regime—there is a clear convergence toward the analytical move of locating accountability in organisational architecture. The convergence operates across common-law and civil-law systems, across primarily civil and primarily criminal accountability regimes, and across different scope catalogues of predicate offences. The shared

analytical move suggests that mediated compliance is not a doctrinally idiosyncratic proposal but rather an articulation, in the international humanitarian law context, of a pattern of legal reasoning that has been independently developed in multiple domestic legal orders.

E. Synthesis of the Comparative Analysis

The four doctrinal analogues examined above—deliberate indifference, Organisationsverschulden, responsabilità amministrativa degli enti, and Article 102 of the Swiss Penal Code—differ substantially in their legal frameworks, their procedural settings, and their sanction regimes. They share, however, a structural recognition that legal accountability may attach to the design of conditions that make harm foreseeable, independent of accountability for the immediate causation of harm.

This shared structural recognition suggests that the analytical move proposed in mediated compliance—locating accountability in the architectural conditions of operational AI systems rather than exclusively in the engagement decision—is not without precedent in legal thought. International Humanitarian Law does not contain a direct equivalent of any of these doctrines. The customary obligations of feasible precaution, distinction, and hors de combat recognition, however, presuppose operational conditions under which compliance is feasible. Where architectural choices render those conditions inoperable, the analytical question of whether and how accountability should attach to the architectural decisions is precisely the question that the comparative doctrines have addressed in domestic contexts.

The comparative analysis does not establish that mediated compliance is binding international law. It establishes that the analytical category is coherent with patterns of legal reasoning that have been developed and refined in other legal orders to address structurally analogous problems.

V. STATE RESPONSIBILITY AND ARCHITECTURAL ATTRIBUTION

The doctrine of state responsibility under international law, codified in the Articles on Responsibility of States for Internationally Wrongful Acts adopted by the International Law Commission in 2001, provides a framework for analysing the attribution of internationally wrongful conduct to a state. The application of this framework to AI-mediated operational systems

raises distinct attribution questions, particularly where the operational architecture is supplied by private contractors and the system is deployed by state actors. This Section examines the implications of mediated compliance for state responsibility analysis.

A. The Standard Attribution Framework

Under Article 4 of the ARSIWA, the conduct of any state organ is attributable to the state. Under Article 5, the conduct of persons or entities empowered by the law of the state to exercise elements of governmental authority is similarly attributable. Under Article 8, the conduct of persons or groups acting on the instructions of, or under the direction or control of, the state is attributable. The standard application of these provisions to military operations conducted by state armed forces is straightforward: the conduct of military personnel constitutes state organ conduct attributable under Article 4.

The introduction of AI-mediated operational systems complicates this attribution analysis in two principal respects. First, where the operational architecture is supplied by private contractors—as is increasingly the case in defence procurement across jurisdictions—the architectural design decisions are made by entities whose relationship to the state is contractual rather than organic. Second, where the architecture itself contributes to operational outcomes through mediation forms of the types described in Section III, the attribution analysis must engage with the question of how design decisions made by non-state actors relate to the conduct of state actors operating within the architectural environment.

B. The Contractor Architecture Problem

Where a private defence contractor supplies operational architecture that subsequently mediates state military operations, two attribution questions arise. First, may the architectural design decisions themselves be attributed to the state, on the basis that the state has effectively delegated the relevant decisions through the procurement process? Second, may the operational consequences of architectural mediation be attributed to the state even where the state actors operating within the architecture acted in good faith on the information presented to them?

The first question engages Article 5 ARSIWA. The procurement of operational architecture for state military use may, depending on the specific procurement framework, constitute an empowerment of the contractor to exercise elements of governmental authority. The provision of

military operational systems is not a routine commercial transaction. Where the state's procurement specifications, acceptance criteria, and deployment authorisations explicitly endorse the architectural choices made by the contractor, the case for Article 5 attribution may be substantial. The architectural decisions are not the contractor's private commercial decisions; they are decisions endorsed by the state and integrated into the state's operational capability.

The second question engages a different analytical structure. State actors operating within an architectural environment that mediates their perception of the operational situation may act in good faith on the information presented to them, while the operational outcome may nonetheless reflect compliance failures attributable to the architectural mediation. The standard attribution rules under Article 4 capture the conduct of the state actor but may not adequately capture the architectural conditions that shaped the conduct. Mediated compliance analysis proposes that the architectural conditions are themselves part of what is attributed to the state, on the basis that the state has authorised the deployment of an architecture whose mediation properties were knowable or should have been knowable at the time of procurement.

C. Due Diligence Obligations

Recent developments in international law have articulated due diligence obligations of states in connection with conduct that occurs within their jurisdiction or under their control. The International Court of Justice in the Corfu Channel case established the principle that a state may not knowingly allow its territory to be used for acts contrary to the rights of other states. The principle has been extended in subsequent jurisprudence and scholarly analysis to encompass a broader range of conduct that the state could have prevented through reasonable diligence.

Applied to AI-mediated operational systems, the due diligence principle suggests that a state procuring operational architecture for military use may incur diligence obligations with respect to the foreseeable compliance-relevant properties of the architecture. Where the procurement authority knew or should have known that the architectural design exhibited mediation properties of the types described in Section III, and where the state nonetheless authorised deployment without addressing those properties, the diligence standard may engage state responsibility independently of the standard attribution framework.

The diligence framework is particularly relevant in connection with dual-use AI systems. Where a system is developed for civilian or general-purpose applications and is subsequently procured for military use, the procurement authority may have access to documentation of the system's architectural properties that the original developers did not generate with military deployment in mind. The diligence inquiry would address whether the procurement authority reasonably evaluated the architectural properties against the operational requirements of the intended military deployment.

D. Implications for Reparation

The forms of reparation available under the ARSIWA—restitution, compensation, satisfaction—may require adaptation to the context of mediated compliance violations. Where the operational consequence of architectural mediation is loss of life or injury to protected persons, restitution in kind is generally impossible, and compensation becomes the principal available remedy. The valuation of compensation for harms attributable to architectural mediation may engage technical questions about the foreseeability of the mediation effects, the available alternative architectural configurations, and the operational pressures that conditioned the procurement decision.

Satisfaction may take particular forms in mediated compliance contexts. Acknowledgement of the architectural inadequacy, public commitment to architectural reform, and adoption of revised procurement standards may constitute satisfaction in ways that are distinctively responsive to the architectural character of the violation. These forms of satisfaction may also have prospective effect, reducing the likelihood of recurrence in a manner that compensation alone does not.

E. The Special Position of Depositary and Host States

Certain states occupy institutional positions under International Humanitarian Law that distinguish them from the general body of states party to the relevant treaties. Among these, Switzerland holds a particularly distinctive position: it is the depositary of the Geneva Conventions of 1949 and their Additional Protocols, the host state of the International Committee of the Red Cross, and the seat of the principal multilateral negotiation processes concerning international

humanitarian law. The institutional position of Switzerland in the international humanitarian law system is unique and is recognised in both treaty practice and customary expectation.

The mediated compliance framework raises a question concerning whether this institutional position carries enhanced due diligence implications for the procurement, hosting, or facilitation of AI-mediated operational systems with potential humanitarian law relevance. The general due diligence obligation discussed in Section V.C applies to all states. The question is whether the depositary and host functions create additional considerations that may modulate the application of that general obligation.

Three considerations inform the analysis. First, the depositary function is principally administrative: it concerns the recording of signatures, ratifications, accessions, and reservations, and the transmission of communications between states parties. The function does not, on its face, generate substantive humanitarian law obligations beyond those imposed by the treaties themselves. The depositary's institutional position is one of trust rather than of substantive normative authority.

Second, the host state function carries additional considerations. Switzerland's role as host of the International Committee of the Red Cross is governed by a headquarters agreement and is reflected in the institutional infrastructure of Geneva, including the broader international humanitarian system. The hosting function generates legitimate expectations on the part of the international community concerning the regulatory environment that Switzerland maintains in connection with activities relevant to humanitarian law. These expectations do not constitute binding legal obligations in any direct sense. They may, however, inform the application of due diligence standards to specific conduct.

Third, and most significantly for the mediated compliance framework, Switzerland's monist legal tradition results in the direct incorporation of ratified humanitarian law treaties and customary international humanitarian law into the domestic legal order. The Federal Council's 2008 implementing legislation, which inserted Articles 264 through 264j into the Penal Code, integrated the core obligations of the Geneva Conventions, the Additional Protocols, and the Rome Statute into Swiss criminal law. The combination of the monist tradition, the comprehensive domestic implementation, and the corporate criminal liability regime under Article 102 produces

a domestic legal infrastructure that is unusually well-positioned to engage architectural mediation as a compliance-relevant phenomenon.

The implications for procurement and hosting decisions are analytically significant. A defence contractor organised under Swiss law, or operating a Swiss subsidiary, that supplies AI-mediated operational architecture to a foreign procuring state, conducts activities that fall within the jurisdiction of Swiss criminal authorities. The architectural mediation forms identified in Section III may, in principle, engage Article 102 paragraph 1 of the Penal Code where the predicate humanitarian law offence is established and the natural person attribution fails for organisational reasons. The Swiss regulatory and prosecutorial framework therefore presents a domestic legal infrastructure within which mediated compliance analysis may be operationalised, independent of the development of corresponding international mechanisms.

This analytical configuration is not a claim that Switzerland is currently exercising such jurisdiction in connection with any specific AI-mediated operational system. It is a structural observation: the combination of Switzerland's institutional position, monist legal tradition, comprehensive humanitarian law implementation, and corporate criminal liability regime produces an unusually favourable analytical environment for the application of mediated compliance principles. The Swiss domestic legal order may therefore serve as a particularly clear illustration of how mediated compliance analysis may be operationalised within existing legal structures, even in the absence of binding international rules specifically governing AI architectural mediation.

More broadly, the analytical attention to depositary and host state positions may inform discussions of how the international humanitarian law system as a whole may evolve to address AI-mediated operational systems. The Geneva-based humanitarian infrastructure—including the ICRC, UNIDIR, the Geneva Academy of International Humanitarian Law and Human Rights, and the Convention on Certain Conventional Weapons Group of Governmental Experts on Lethal Autonomous Weapons Systems—provides an institutional setting in which the analytical resources for engaging architectural mediation may be developed and disseminated. The mediated compliance framework proposed in this paper is offered, in part, as a contribution to that institutional process.

VI. OPERATIONAL AND PROCUREMENT IMPLICATIONS

The analytical framework developed in the preceding sections has practical implications for the design of procurement, audit, and operational accountability processes in connection with AI-mediated operational systems. This Section identifies four such implications.

A. Architectural Specification as a Procurement Document

Procurement specifications for AI-mediated operational systems may, in light of mediated compliance analysis, need to address architectural mediation forms explicitly. Specifications that address only system performance against threat detection benchmarks may capture insufficient information about the system's mediation properties. Specifications that additionally address the system's behaviour with respect to compliance-relevant signal classes—surrender indicators, hors de combat configurations, ambiguity preservation pathways—may provide a more complete basis for procurement decisions that engage with mediated compliance considerations.

The Italian regime under Legislative Decree 231/2001 examined in Section IV.C provides a structural model for the documentation of architectural specifications. The organisational model under that regime must be written, verifiable, and subject to independent supervision. Procurement specifications for operational AI systems may adopt a similar documentary structure: written specifications of architectural mediation properties, verifiable through independent testing, and subject to documented review by appropriate authorities.

B. Independent Pre-Deployment Audit

Independent audit of operational AI systems prior to deployment may, in light of mediated compliance analysis, need to engage the architectural mediation taxonomy proposed in Section III. An audit that examines sensor specifications, classification taxonomies, threshold calibrations, and priority weighting structures may identify compliance-relevant mediation properties that would not be detected by performance-focused testing alone.

The independence of the audit is particularly significant. Where the audit is conducted by entities affiliated with the system developer or the procurement authority, the structural conditions for independent assessment may be compromised. Mediated compliance analysis suggests that the

credibility of pre-deployment audit may depend substantially on the institutional independence of the auditing entity from the parties whose architectural decisions are being assessed.

C. Command Awareness and the "Should Have Known" Standard

Article 28 of the Rome Statute of the International Criminal Court establishes that a military commander may incur individual criminal responsibility where the commander knew or, owing to the circumstances at the time, should have known that the forces under their effective command and control were committing or about to commit crimes within the jurisdiction of the Court. The "should have known" standard is objective: it does not require subjective awareness on the part of the commander but only that the commander was in a position where awareness was reasonably to be expected.

Mediated compliance analysis has implications for the application of the "should have known" standard in AI-mediated operational contexts. As architectural mediation properties of operational AI systems become documented in procurement specifications, audit reports, and scholarly analysis, the objective conditions for commander awareness are progressively established. A commander authorising deployment of a system whose architectural mediation properties are documented in publicly available materials may, depending on the specific factual context, be in a position where the standard's awareness threshold is engaged.

This implication is not a claim that command responsibility automatically attaches in connection with any AI-mediated operational deployment. It is a more limited observation: that as the analytical framework of mediated compliance becomes established in scholarly and procurement discourse, the objective conditions under which commander awareness is reasonably to be expected may shift. The progressive accumulation of analytical materials about architectural mediation contributes to defining the "should have known" baseline for future operational deployments.

D. Documentation as Compliance Architecture

Across the procurement, audit, and command awareness implications discussed above, a common feature emerges: documentation functions as the operational substrate of mediated compliance analysis. Written architectural specifications, recorded audit findings, documented feasibility assessments, and acknowledged command awareness records together constitute a

documentary architecture that may be reviewed retrospectively in connection with operational outcomes.

This documentary architecture serves two functions. First, it operationalises the analytical framework of mediated compliance by providing concrete artefacts against which accountability analysis may be conducted. Second, it has prospective effect: the requirement to document architectural choices may itself influence the choices that are made, by surfacing considerations that informal procurement processes might overlook.

Documentary architecture is not a substitute for substantive compliance. It is, however, an operational mechanism through which substantive compliance may be assessed, contested, and improved. The development of standard documentary frameworks for AI-mediated operational systems—building on existing initiatives such as the NIST AI Risk Management Framework, IEEE 7000-2021, and NATO AI assurance protocols—may constitute a significant practical implementation of mediated compliance analysis.

VII. LIMITATIONS OF THE ARGUMENT

The analytical framework developed in this paper is subject to several limitations that warrant explicit acknowledgement.

First, the framework does not establish that mediated compliance is binding international law. The argument is descriptive-analytical: it proposes that the existing framework of humanitarian and state responsibility law may contain the analytical resources necessary to address architectural mediation. Whether and how these resources are deployed in specific cases is a question of legal practice and adjudication that this paper does not resolve.

Second, the taxonomy of architectural mediation forms proposed in Section III is provisional. The four categories—sensor filtering, taxonomic exclusion, threshold calibration, and priority weighting—are intended as analytical reference points rather than exhaustive descriptions. Operational AI systems may exhibit mediation forms not captured by this taxonomy, and the boundaries between categories may not always be sharp. Refinement of the taxonomy through application to specific systems is an open research agenda.

Third, the comparative analysis in Section IV does not establish that the domestic doctrines examined are directly transposable to International Humanitarian Law. The doctrinal analogues are offered as evidence that the analytical move proposed in mediated compliance has precedent in legal reasoning, not as a claim that the doctrines themselves apply at the international level. The transposition of analytical structures across legal orders is a delicate operation that requires attention to differences in procedural setting, sanction regime, and institutional context.

Fourth, the state responsibility analysis in Section V depends on contestable interpretations of the ARSIWA framework. The application of Article 5 attribution to defence contractors supplying operational architecture, and the extension of due diligence obligations to AI procurement decisions, are matters on which the existing case law and scholarly literature provide only partial guidance. The analysis proposed in this paper represents a possible application of the framework rather than a settled interpretation.

Fifth, the operational and procurement implications discussed in Section VI assume institutional conditions—independent audit capacity, documentary discipline, command awareness practices—that may not be uniformly available across jurisdictions and operational contexts. The practical implementation of mediated compliance analysis will depend on the institutional infrastructure available in specific settings.

Finally, the framework is silent on a range of related questions that warrant separate analysis: the application of mediated compliance to non-international armed conflicts, the relationship between mediated compliance and the law of state immunity in connection with operational systems supplied across borders, and the implications for individual criminal responsibility under the Rome Statute beyond the command responsibility doctrine. Each of these questions may be approached using the analytical resources developed here but requires independent elaboration that this paper does not undertake.

VIII. CONCLUSION

The legal analysis of AI-mediated operational systems may benefit from explicit recognition of the analytical category proposed in this paper. Mediated compliance—the condition under which the practical operability of a legal obligation depends on technical infrastructure designed and procured by actors distinct from the human decision-maker who bears the formal

obligation—identifies a phenomenon that existing analytical frameworks address only partially. The phenomenon is not new: organisational and architectural conditions have always shaped the operational environment in which legal obligations are discharged. What is new is the degree to which algorithmic mediation has become constitutive of operational perception and decision in military-adjacent environments, and the consequent salience of architectural decisions for compliance analysis.

The framework proposed here does not resolve the difficult questions that arise at the intersection of AI systems and International Humanitarian Law. It proposes a distinction—between architectural conditions that condition compliance and decisions that discharge compliance—that may clarify the structure of those questions. Under the proposed framework, accountability analysis may need to engage with the full distribution of decision points across the procurement and deployment chain, including architectural design, training, calibration, and procurement decisions that precede the engagement moment.

The comparative analysis with deliberate indifference, *Organisationsverschulden*, *responsabilità amministrativa degli enti*, and Article 102 of the Swiss Penal Code suggests that the analytical move proposed here is coherent with patterns of legal reasoning developed in domestic legal orders to address structurally analogous problems. The convergence across common-law and civil-law systems, and across primarily civil and primarily criminal accountability regimes, suggests that mediated compliance is not a doctrinally idiosyncratic proposal but rather an articulation of a broader pattern of legal reasoning, in the specific context of International Humanitarian Law. The translation of these patterns to the international level is not automatic, but the analytical resources for the translation are available, and the Swiss domestic legal infrastructure may provide a particularly clear setting in which the framework may be operationalised.

The implications for state responsibility, command responsibility, and procurement practice are significant. As AI-mediated operational systems become more deeply integrated into military and military-adjacent environments, the architectural decisions that condition their operational behaviour may acquire correspondingly greater legal salience. The development of documentary frameworks that surface architectural mediation properties, of audit practices that engage with the mediation taxonomy proposed here, and of command awareness practices that

address the objective conditions for the "should have known" standard, may together constitute a practical implementation of mediated compliance analysis.

The paper does not claim that the development of these practices is mandated by current international law. It proposes that the analytical resources for such development are available within the existing framework of humanitarian and state responsibility law, and that articulating mediated compliance as a distinct analytical category may clarify ongoing scholarly and policy discussions concerning the legal accountability of AI-mediated operational systems.

METHODOLOGICAL NOTE

This paper proposes an analytical framework rather than a positive statement of existing customary obligations concerning AI architecture or state responsibility. The argument is offered as a contribution to ongoing scholarly and policy discussion. It does not purport to resolve disputed questions of treaty interpretation or customary law formation, nor to establish binding standards of conduct.

The taxonomy of architectural mediation forms presented in Section III, the comparative doctrinal analysis in Section IV, and the state responsibility analysis in Section V are offered as analytical constructs. They are intended to demonstrate how the proposed framework may apply to operational configurations and legal questions, not to describe or evaluate any existing system or to resolve any specific legal dispute.

The operational and procurement implications discussed in Section VI are offered as analytical reference points and do not constitute legal advice. The framework is presented for engagement, critique, and further development by scholars, practitioners, and policymakers working in international humanitarian law, state responsibility, autonomous systems governance, and defence procurement.

This paper develops a concept introduced in summary form in the author's earlier work on lawful operational safeguards. The present paper extends the analytical foundation, develops the taxonomy of mediation forms, conducts the comparative doctrinal analysis, and addresses state responsibility implications that the earlier work identified but did not fully elaborate.

Cite as: Giovanni Nardacci, 'Mediated Compliance: Architectural Constraints on the Operability of International Humanitarian Law Obligations in AI-Mediated Operational Environments' (2026) (developing the analytical category of mediated compliance within existing International Humanitarian Law and state responsibility frameworks).

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